

Graphs and signed graphs with integer reciprocal eigenvalue properties

Callum Huntington

University of Naples Federico II, Italy

2nd Joint Meeting Brazil–Italy in Mathematics
Messina, Italy

Joint work with Francesco Belardo (University of Naples Federico II, Italy)
and Sasmita Barik (IIT Bhubaneswar, India).

Dedicated to Professor Vilmar Trevisan.

The properties

Reciprocal eigenvalue properties

The reciprocal eigenvalue property was first introduced by Barik, Neumann and Pati in 2006.

Barik, Pati and Verma later introduced the integer-valued version.

Definition

Let m be a positive integer and let (Γ, σ) be a signed graph. Then (Γ, σ) satisfies

- m -(R) if λ an eigenvalue implies m/λ an eigenvalue, for every $\lambda \neq 0$;
- m -(−R) if λ an eigenvalue implies $-m/\lambda$ an eigenvalue, for every $\lambda \neq 0$.

If the paired eigenvalues always have equal multiplicity, then (Γ, σ) satisfies m -(SR), respectively m -(−SR).

Setting $m = 1$ recovers the classical (R), (SR), (−R) and (−SR).

Symmetric spectrum

An ordinary, unsigned graph Γ is treated as the all-positive signed graph $(\Gamma, +)$.

Definition

A signed graph (Γ, σ) has *symmetric spectrum* if λ and $-\lambda$ occur with equal multiplicity for every λ .

For unsigned graphs this is equivalent to bipartiteness.

For signed graphs it is strictly weaker: there are non-bipartite signed graphs with symmetric spectrum.

Products, coronae, hypercubes

Kronecker products

Following Germina, Hameed and Zaslavsky, the Kronecker product $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ has vertex set $V_{\Gamma_1} \times V_{\Gamma_2}$, with $(u_i, v_j) \sim (u_r, v_s)$ when $u_i \sim u_r$ and $v_j \sim v_s$, taking the signature $\sigma_1(u_i u_r) \sigma_2(v_j v_s)$.

Its adjacency matrix is $A_{(\Gamma_1, \sigma_1)} \otimes A_{(\Gamma_2, \sigma_2)}$, so the eigenvalues are the products $\lambda_i \mu_j$.

Proposition

If (Γ_1, σ_1) satisfies m_1 -(R) and (Γ_2, σ_2) satisfies m_2 -(R), then $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ satisfies $m_1 m_2$ -(R).

The proof is a one-line computation: $(m_1/\lambda_i)(m_2/\mu_j) = m_1 m_2 / (\lambda_i \mu_j)$.

Corollary

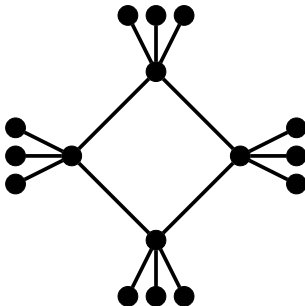
With an abuse of notation... m_1 -(R) $\otimes$ m_2 -($-R$) gives $m_1 m_2$ -($-R$), and m_1 -($-R$) $\otimes$ m_2 -($-R$) gives $m_1 m_2$ -(R).

The strong versions survive too, pairings in each factor induce a multiplicity-preserving bijection, so repeated products are fine.

Signed simple m -coronae

The simple m -corona of a graph Γ , denoted by $\Gamma \circ \bigcup_m K_1$, attaches m pendant vertices to every vertex of Γ .

For example, here we have the simple 3-corona of C_4 :



Signed simple m -coronae proposition

Proposition

Let (Γ, σ) be a signed graph on $n \geq 2$ vertices and let m be a positive integer. Then $(\Gamma, \sigma) \circ \cup_m(K_1, +)$ satisfies m -(−SR). Furthermore, if (Γ, σ) has symmetric spectrum, then it also satisfies m -(SR).

Note that m -(−SR) holds for every signed root graph on $n \geq 2$ vertices.

Compare Barik–Pati–Verma, who obtain m -(−R) in general and both strong properties for bipartite roots. So this strengthens the conclusion and weakens the hypothesis simultaneously.

Signed simple m -coronae: the argument (1/2)

With a suitable ordering of the vertices,

$$A_{((\Gamma, \sigma) \circ \cup_m(K_1, +))} = \begin{pmatrix} A_{(\Gamma, \sigma)} & I_n \otimes \mathbf{1}_m^T \\ I_n \otimes \mathbf{1}_m & O_{mn} \end{pmatrix},$$

applying Schur's complement formula gives, for $\lambda \neq 0$,

$$P((\Gamma, \sigma) \circ \cup_m(K_1, +); \lambda) = \lambda^{mn} P\left((\Gamma, \sigma); \lambda - \frac{m}{\lambda}\right).$$

So non-zero λ is an eigenvalue of the simple m -corona precisely when $\lambda - m/\lambda$ is an eigenvalue of the root graph.

Signed simple m -coronae: the argument (2/2)

Solving $\lambda^2 - \mu\lambda - m = 0$ for each root eigenvalue μ gives

$$\lambda^{\pm} = \frac{\mu \pm \sqrt{\mu^2 + 4m}}{2}, \quad \lambda^+ \lambda^- = -m.$$

Hence $\lambda^- = -m/\lambda^+$. Both arise from the same μ , so their multiplicities match, giving m -(−SR).

If the spectrum is symmetric, $-\mu$ contributes $-\lambda^{\mp}$ and $m/\lambda^{\pm} = -\lambda^{\mp}$ gives m -(SR).

The identity divides by λ , so the nullity is found separately. Each root vertex contributes $m - 1$ pendant vectors summing to zero, giving a nullity of $n(m - 1)$ whatever the root.

Hypercubes

The hypercube Q_n has vertex set $\{0, 1\}^n$, with two vertices adjacent when they differ in exactly one coordinate. It is n -regular on 2^n vertices.

Q_n has spectrum $\{(n - 2k) \binom{n}{k} : k = 0, 1, \dots, n\}$, which is integral.

So,

$$\text{spec}(Q_1, +) = \{1, -1\},$$

$$\text{spec}(Q_2, +) = \{2, 0^{(2)}, -2\},$$

$$\text{spec}(Q_3, +) = \{3, 1^{(3)}, -1^{(3)}, -3\},$$

$$\text{spec}(Q_4, +) = \{4, 2^{(4)}, 0^{(6)}, -2^{(4)}, -4\},$$

and Q_1 satisfies 1-(SR), Q_2 satisfies 4-(SR), Q_3 satisfies 3-(R), and Q_4 satisfies 8-(R), each with its anti-reciprocal counterpart.

Hypercubes proposition

Proposition

Let $n \geq 1$. There is a positive integer m with Q_n satisfying m -(R) if and only if $n \leq 4$. The same holds for m -(-R).

Proof.

We handled $n \leq 4$ on the previous slide.

Suppose $n \geq 5$. The spectrum is integral, so every non-zero eigenvalue must divide m , and hence so does their lowest common multiple.

Since n is the largest eigenvalue and m/n must also be an eigenvalue, $m/n \leq n$, forcing $m \leq n^2$.

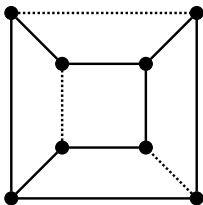
But $\text{lcm}(n, n-2, n-4)$ grows cubically and outruns n^2 .

Only $n \in \{5, 6, 8, 12\}$ need checking by hand, and each fails. □

Signed hypercubes

Huang's 2019 proof of the sensitivity conjecture uses a signing of the hypercube Q_n .

Laplante, Naserasr, Sunny and Wang recorded in 2026 that for each $n \geq 2$ there is a signature σ_n making every 4-cycle of Q_n negative.



Such a signed hypercube has two distinct eigenvalues, $\sqrt{n}$ and $-\sqrt{n}$, each with multiplicity 2^{n-1} . This is equivalent to $A^2_{(Q_n, \sigma_n)} = nI$.

Signed hypercubes proposition

Proposition

For $n \geq 2$, the signed hypercube (Q_n, σ_n) in which all 4-cycles are negative satisfies both n -(SR) and n -(-SR).

So the unsigned family fails from $n = 5$ onwards, while the signed family satisfies the properties for every $n \geq 2$, and satisfies the strong versions rather than the weak ones.

Introducing a (suitable) signature therefore gives properties that the unsigned hypercube cannot have for any $n \geq 5$.

Graphs satisfying m -(R) but not m -(SR)

Separating m -(R) from m -(SR)

Clearly m -(SR) implies m -(R). The converse fails, and we construct an infinite family demonstrating this, generalising a construction of Panda and Pati.

Construction

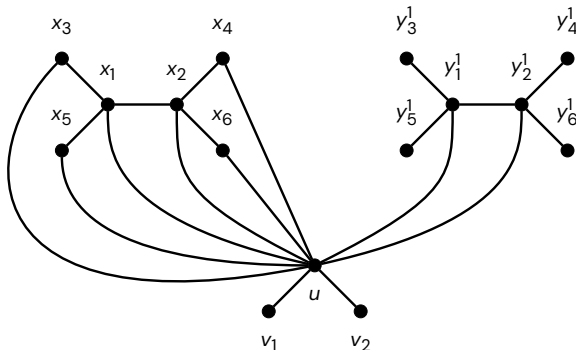
Set $R = R_1 \circ \cup_m K_1$ with $R_1 = K_1$, $V_{R_1} = \{u\}$. Let $\Gamma_{m,n}$ with $m \geq 1$ and $n \geq 2$, be obtained by taking n copies of $K_2 \circ \cup_m K_1$ and one copy of R , then joining u to every vertex of the first copy of $K_2 \circ \cup_m K_1$ and to the vertices of degree $m + 1$ in the remaining $n - 1$ copies.

Write

$$\Omega_1^m = \{\Gamma_{m,n} : m, n \in \mathbb{N}, m \geq 1, n \geq 2\}.$$

Setting $m = 1$ recovers the Panda–Pati (2016) construction exactly.

A small example



The graph $\Gamma_{2,2} \in \Omega_1^2$, which satisfies 2-(R) but not 2-(SR).

The characteristic polynomial

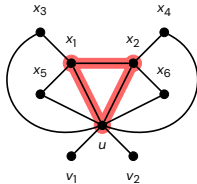
Theorem

Let $\Gamma_{m,n} \in \Omega_1^m$. Then $\Gamma_{m,n}$ satisfies m -(R) but not m -(SR), and

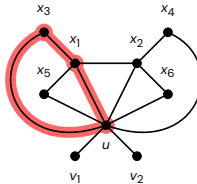
$$P(\Gamma_{m,n}; \lambda) = \underbrace{(\lambda^{2m+2} - (2m+1)\lambda^{2m} + m^2\lambda^{2m-2})^{n-1}}_{\text{from the } n-1 \text{ outlying copies}} \\ \times \left\{ \lambda^{3m+3} - (2n+5m+1)\lambda^{3m+1} - (2n+4m)\lambda^{3m} \right. \\ \left. + m(2n+5m-1)\lambda^{3m-1} + 2m^2\lambda^{3m-2} - m^3\lambda^{3m-3} \right\}.$$

The computation runs on two classical formulas of Schwenk: one stripping away pendant vertices, and one expanding about u over all cycles through it.

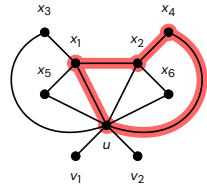
Every cycle of $\Gamma_{m,n} - \{v_1, \dots, v_m\}$ passes through u , and these cycles fall into six types, counted 1, $2m$, $2m$, m^2 , $m(m-1)$, and $n-1$ times respectively.



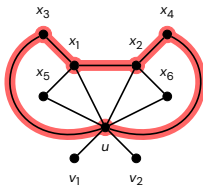
Type (i)



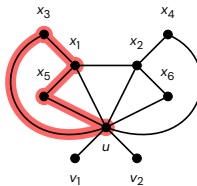
Type (ii)



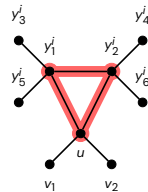
Type (iii)



Type (iv)



Type (v)



Type (vi)

Why m -(R) holds and m -(SR) fails

Write the two bracketed factors of $P(\Gamma_{m,n}; \lambda)$ as $F_1(\lambda)$ and $F_2(\lambda)$.
Their roots are

$$\lambda_{(1,\pm)} = \frac{1}{2}(1 \pm \sqrt{4m+1}), \quad \lambda_{(2,\pm)} = \frac{1}{2}(-1 \pm \sqrt{4m+1}),$$

from F_1 , together with $\lambda_{(3,\pm)} = \lambda_{(2,\pm)}$ and two further pairs $\lambda_{(4,\pm)}, \lambda_{(5,\pm)}$ from F_2 .

Every one of these pairs multiplies to m , so each non-zero eigenvalue is paired with its m -reciprocal and m -(R) holds.

The failure is one of multiplicity. Since $\lambda_{(2,\pm)} = \lambda_{(3,\pm)}$, these are roots of *both* F_1 and F_2 , while $\lambda_{(1,\pm)}$ is a root of F_1 only. Hence

$$\text{mult}(\lambda_{(2,\pm)}) = n \quad \text{but} \quad \text{mult}(\lambda_{(1,\pm)}) = n - 1,$$

and m -(SR) fails.

A signed variant

Repeat the construction but this time take every edge negative.

Since each $\Gamma_{m,n}$ contains triangles, such as ux_1x_2 , the signed graph $(\Gamma_{m,n}, -)$ is always unbalanced.

As $A_{(\Gamma,-)} = -A_{\Gamma}$, the spectrum is negated and the conclusion is immediate: $(\Gamma_{m,n}, -)$ also satisfies m -(R) but not m -(SR).

Non-corona graphs satisfying m -(SR) and m -(—SR)

Non-corona constructions

Simple m -coronae give m -(-SR) and often m -(SR), as we saw. However the corona structure is not a necessity.

We construct an infinite family of graphs which satisfy both strong properties and are *not* simple m' -coronae for any $m' \geq 1$.

Construction

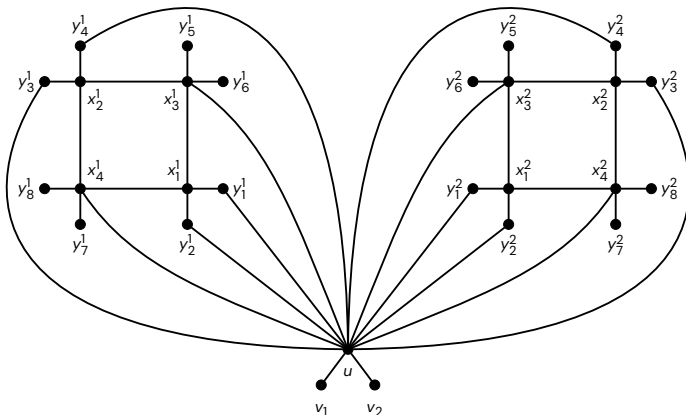
Let $\Gamma_{k,m,n}$ with $k \geq 2$ and $m, n \geq 1$ be built as follows. Begin with a vertex u and attach pendants $v_1, \dots, v_m$. Take n copies of C_{2k} , labelled bipartitely so that $x_{k+i} \sim x_i$ and $x_{k+i} \sim x_{i+1}$ for $1 \leq i \leq k$, indices modulo k . Form the simple m -corona of each copy. Join u to the first km pendant vertices and to the last k cyclic vertices of each copy.

Write

$$\Omega_2^m = \{\Gamma_{k,m,n} : k, m, n \in \mathbb{N}, k \geq 2, m, n \geq 1\}.$$

Setting $m = 1$ recovers the class $\mathcal{NC}_1^1$ of Rakshith, Das and Manjunatha (2022).

The family Ω_2^m



The non-corona graph $\Gamma_{2,2,2} \in \Omega_2^2$, which satisfies 2-(SR) and 2-(SR).

Theorem and sketch proof

Theorem

Let $\Gamma_{k,m,n} \in \Omega_2^m$. Then $\Gamma_{k,m,n}$ satisfies both m -(SR) and m -(-SR).

Write $\begin{pmatrix} O_k & C \\ C^T & O_k \end{pmatrix}$ for the adjacency matrix of C_{2k} , and let λ_i be the singular values of C . Eigenvectors built from the singular vector pairs of C contribute

$$c_i^\pm = \frac{\lambda_i \pm \sqrt{\lambda_i^2 + 4m}}{2}, \quad d_i^\pm = \frac{-\lambda_i \pm \sqrt{\lambda_i^2 + 4m}}{2},$$

with $c_i^+ c_i^- = -m$ and $c_i^+ d_i^+ = m$, giving both pairings at once.

Pendant vectors summing to zero give 0 with multiplicity $(2kn + 1)(m - 1)$.

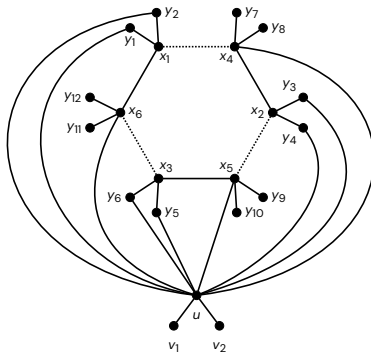
An equitable partition into six cells accounts for the remaining eigenvalues, which satisfy

$$(\mu^2 - m)(\mu^4 - (kn(m + 1) + 2m + 4)\mu^2 + m^2) = 0.$$

The constant term is m^2 , so these pair correctly too.

The signed variant

Let (C_{2k}, τ) be the signed cycle in which τ is -1 on the perfect matching $\{\{x_i, x_{k+i}\}\}$ and $+1$ on the complementary matching. Take all remaining edges positive.



$(\Gamma_{3,2,1}, \sigma) \in (\Omega_2^2, \Sigma)$: non-corona, unbalanced, satisfying 2-(SR) and 2-(SR).

What changes, and what does not

The underlying graph is unchanged, so the signed graph is also not a simple corona.

Unbalancedness holds in every case: the cycle $u y_1^1 x_1^1 x_{k+1}^1 u$ carries exactly one negative edge.

Theorem

Let $(\Gamma_{k,m,n}, \sigma) \in (\Omega_2^m, \Sigma)$. Then $(\Gamma_{k,m,n}, \sigma)$ satisfies m -(SR) and m -(SR).

The proof follows the same route, with one structural change: the cycle is 0-net-regular rather than 2-regular, so $C1_k = 0$ instead of $2 \cdot 1_k$, giving $\lambda_1 = 0$. This propagates in two visible places.

- $c_1^\pm = d_1^\pm = \pm\sqrt{m}$, with $2(n-1)$ copies, in place of $\pm 1 \pm \sqrt{1+m}$.
- The coefficient $kn(m+1) + 2m + 4$ becomes $kn(m+1) + 2m$, so

$$(\mu^2 - m)(\mu^4 - (kn(m+1) + 2m)\mu^2 + m^2) = 0.$$

The constant term is still m^2 and the pairing survives untouched.

Summary

- We introduced the integer anti-reciprocal properties m -(−R) and m -(−SR).
- All four properties interact cleanly with the Kronecker product, multiplying the parameters.
- Signed simple m -coronae always satisfy m -(−SR), with m -(SR) being satisfied whenever the root has symmetric spectrum.
- The unsigned hypercube Q_n satisfies an integer reciprocal property if and only if $n \leq 4$, and a signing to satisfy n -(SR) and n -(−SR) for every $n \geq 2$ exists.
- Ω_1^m and its signed variant separate m -(R) from m -(SR), by a multiplicity mismatch of exactly one.
- Ω_2^m gives non-corona graphs with both strong properties, and admits an unbalanced signed variant.

Thank you!

Grazie!

Obrigado!